DESCRIPTIVE STATISTICS · STUDY NOTES

Correlation Coefficient

Covariance made easy to read: one number between −1 and 1

30-SECOND REVISION

Correlation coefficient r = covariance ÷ (SD of x × SD of y). It fixes covariance's problem: **no units, always between −1 and 1**.
+1 perfect positive · **between 0 and 1** positive · **0** no linear relationship · **between −1 and 0** negative · **−1** perfect negative.
Examples from class: house size vs price r = **0.87**; SAT writing vs reading r = **0.94**.
Remember: r is symmetric (x with y = y with x), and **correlation does not imply causation**.

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1. From covariance to correlation

The previous topic ended with one problem: a covariance such as **33,491,250** tells you the **direction** of a relationship (positive), but you cannot tell whether it is strong or weak. The **correlation coefficient** adjusts the covariance so the relationship becomes easy to interpret.

	Covariance	Correlation coefficient
Tells you the direction	Yes (by its sign)	Yes (by its sign)
Tells you the strength	No	**Yes**
Units	unit of x × unit of y	**None**
Range	any number	**−1 to 1**
Easy to compare across data sets	No	**Yes**

2. What the correlation coefficient is

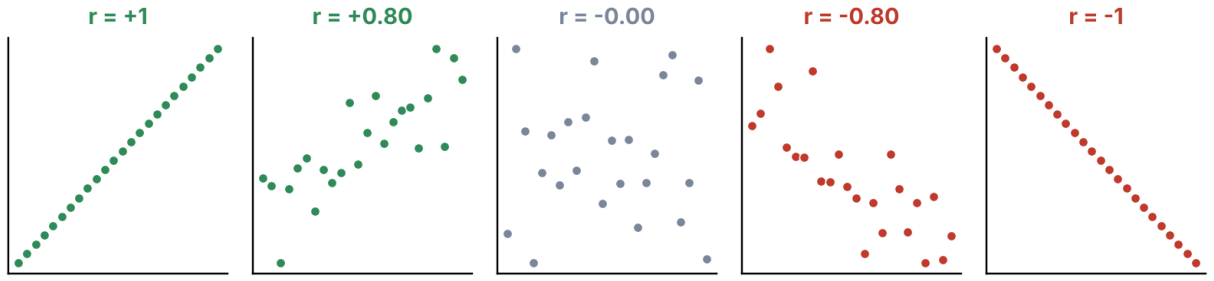
Definition. The correlation coefficient is the covariance divided by the standard deviations of the two variables. Use the sample or the population versions consistently: if the covariance and the standard deviations are all sample values, the result is the sample correlation r; if all are population values, it is ρ (rho). The numerical result is the same either way, because the n − 1 or N cancels.

$$r = \frac{\operatorname{cov}(x, y)}{s_x s_y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})/(n-1)}{s_x s_y}$$

$$\rho = \frac{\operatorname{cov}(x, y)}{\sigma_x \sigma_y}$$

How to read the value

r	Name	Meaning
+1	Perfect positive correlation	All points lie exactly on a rising straight line
between 0 and +1	Imperfect positive	x and y tend to rise together; closer to 1 is stronger
0	No linear correlation	No straight-line relationship between x and y
between −1 and 0	Imperfect negative	One tends to rise as the other falls; closer to −1 is stronger
−1	Perfect negative correlation	All points lie exactly on a falling straight line



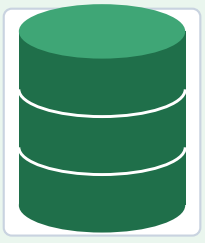
Illustrative scatter plots (not class data) showing how r changes with the pattern of the points.

Mathematical fact
The correlation coefficient can **never** be greater than 1 or less than −1. If your calculation gives 1.3 or −2, there is an error.

3. Worked example: housing data

The same five houses as in the covariance topic. Because they are only part of all houses, this is a **sample**.

3.1 The data set



click to open

Housing: Size vs Price

DATA SET EXAMPLE

Data set example · 25.0 KB · 3 sheets

Housing data: 5 houses with their size and price

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

House	Size (ft.)	Price ($)
1	650	$ 772,000
2	785	$ 998,000
3	1,200	$ 1,200,000
4	720	$ 800,000
5	975	$ 895,000

3.2 The steps

Step 1: covariance. From the previous topic: cov(size, price) = 133,965,000 ÷ 4 = **33,491,250**.

Step 2: standard deviation of size. Sum of squared distances = 197,970. Divide by 4 to get the variance 49,492.50, then take the square root: s_x = **222.47** ft.

Step 3: standard deviation of price. Sum of squared distances = 120,568,000,000. Divide by 4 to get the variance 30,142,000,000, then take the square root: s_y = **$ 173,614.52**.

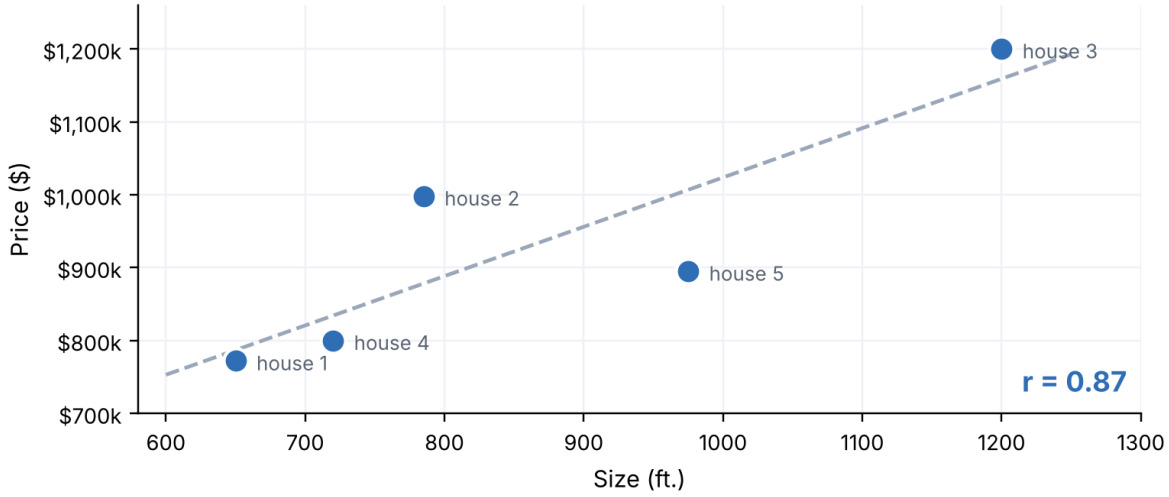
Step 4: divide. r = 33,491,250 ÷ (222.47 × 173,614.52) = **0.8671** ≈ **0.87**.

The building blocks

House	x − 866	$(x - \bar{x})^2$	y − 933,000	$(y - \bar{y})^2$ (billions)	$(x - \bar{x})(y - \bar{y})$
1	-216	46,656	-161,000	25.921	+34,776,000
2	-81	6,561	+65,000	4.225	-5,265,000
3	+334	111,556	+267,000	71.289	+89,178,000
4	-146	21,316	-133,000	17.689	+19,418,000
5	+109	11,881	-38,000	1.444	-4,142,000
Sum		**197,970**		**120.568**	**133,965,000**

$$r = \frac{33{,}491{,}250}{222.47 \times 173{,}614.52} = 0.87$$

What the plot shows



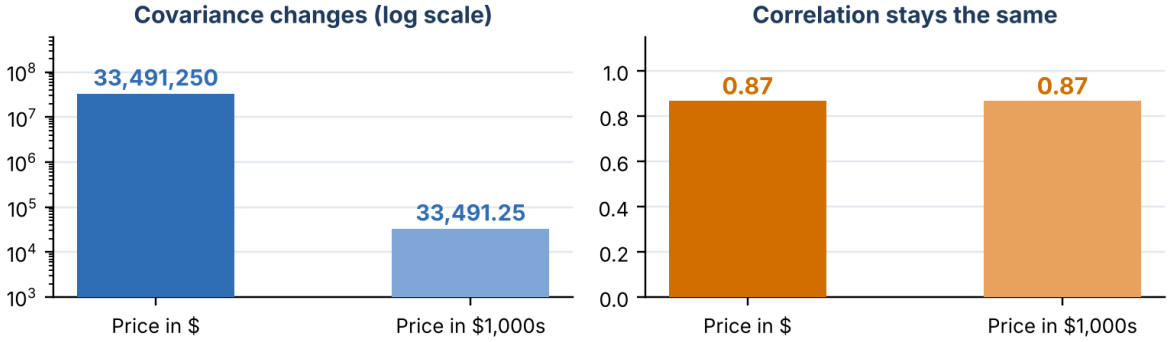
The points follow a rising line closely, which is why r is high. The dashed line is the best-fit straight line (added for illustration).

Result
r = **0.87**. The correlation is **strong and positive**: larger houses tend to cost more.
Careful: r is a pure number. It does **not** mean that each extra foot adds $870 to the price. The price gained per foot is the slope of the fitted line, cov ÷ variance of size = 33,491,250 ÷ 49,492.50 ≈ **$ 677** per ft. (added; see the notes check).

How much does size explain? (added)
Squaring r gives $r^2 = 0.87^2 \approx$ **0.75**: about **75%** of the variation in price goes along with size in a straight line. The other 25% is other factors (location, age, condition). So size is the main driver here, but not the only one.

4. Properties of the correlation coefficient

- **Between −1 and 1.** Always. This is what makes it easy to interpret.
- **Unit-free.** The units of x and y cancel when you divide the covariance by the two standard deviations.
- **Symmetric.** The correlation between x and y equals the correlation between y and x. The formula is completely symmetric: swapping the variables gives 0.87 again.
- **Not changed by rescaling.** Change dollars to thousands of dollars, or feet to metres, and r stays the same. The covariance does not. (The check below was calculated from the housing data.)



Perfect correlation, in numbers (added)

x	y	Pattern	r
1, 2, 3	2, 4, 6	y rises by 2 for every 1 of x	**+1**
1, 2, 3	6, 4, 2	y falls by 2 for every 1 of x	**−1**

r = 1 in the real estate context
The class example: if every extra square foot added the same amount to the price (say 1 sq. ft. = $1,000, so 5 sq. ft. = $5,000), all houses would lie on one straight line and r would be exactly **1**. Real data never behaves that perfectly, which is why r = 0.87 is not 1.

5. Correlation and causation

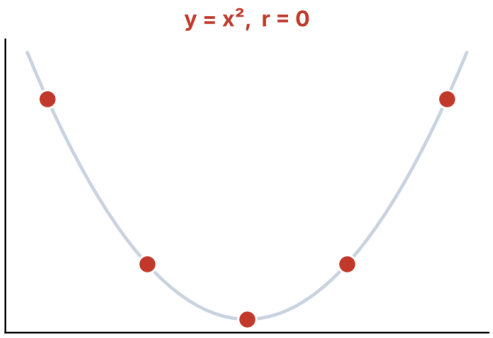
Correlation measures **whether** two variables move together. It says nothing about **why**.

Idea	What the class notes say
Direction of cause	In the house business, **size causes price**, not the other way round: you decide the size when you build, and a bigger house costs more. Raising the price of an existing house does not make it bigger.
Why it matters	Causality tells the direction of a causal relationship. Correlation is symmetric, so it cannot tell you which variable is the cause.
Conclusion	**Correlation does not imply causation.**

Zero and negative correlation (class examples)

Case	Example	What happens
r = 0	Coffee production in Brazil and house prices in London	The two variables have nothing in common, so no relationship shows up.
r < 0	Ice-cream sales and umbrella sales	Ice cream sells more in good weather, umbrellas sell more when it rains: when one rises, the other falls.
r = −1	Perfect negative correlation	A falling straight line
−1 < r < 0	Imperfect negative correlation	Falling pattern with scatter, like the ice-cream and umbrella case

Two warnings (added)
1. r = 0 means no *linear* relationship, not "no relationship". The points x = −2, −1, 0, 1, 2 and y = 4, 1, 0, 1, 4 follow $y = x^2$ perfectly, yet r = 0 (right).
2. A strong r can come from a third factor. Ice cream and umbrellas are both driven by the weather, not by each other. Always ask what could explain the link before claiming a cause.

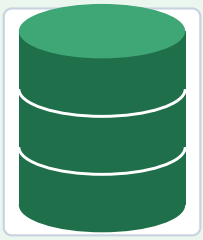


6. Exercise: SAT scores

Data on SAT writing and reading scores from the earlier correlation exercise. The same five students as in the covariance exercise.

Task	What to do	Answer
1	Calculate the correlation coefficient of the two data sets	**r = 0.94**
2	Comment on the strength of the correlation	**Very strong positive** (0.94 is extremely close to 1)

6.1 The data set



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SAT Scores: Writing vs Reading

DATA SET
EXAMPLE

Data set example · 29.5 KB · 3 sheets

Exercise and solution: 5 students, SAT writing and reading

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

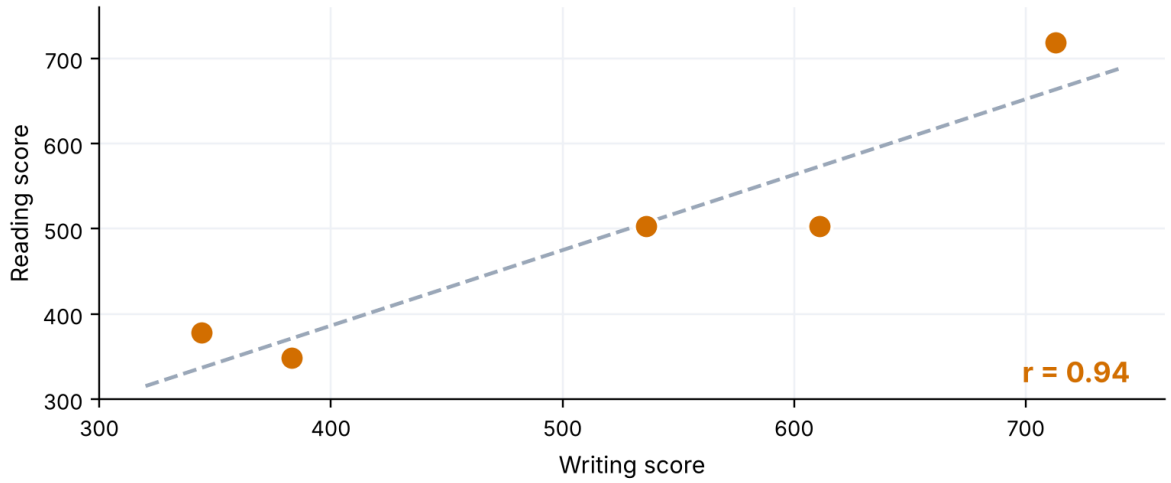
Student	Writing (x)	Reading (y)
1	344	378
2	383	349
3	611	503
4	713	719
5	536	503

6.2 Solution, step by step

Quantity	Writing (x)	Reading (y)
Mean	517.4	490.4
Variance (÷ 4)	23,874.3	21,300.8
Standard deviation	154.51	145.95
Covariance (÷ 4)	21,155.55	

$$r = \frac{21{,}155.55}{154.51 \times 145.95} = 0.94$$

Task 2. r = 0.94 is extremely close to 1, so there is a **very strong positive correlation**: students with higher writing scores almost always have higher reading scores. As in the class solution, the covariance in the workbook is calculated by hand, and the coefficient with the built-in CORREL function; both approaches agree.



Compare the two examples (added)
The SAT correlation (**0.94**) is higher than the housing one (**0.87**), so the SAT relationship is the stronger one, even though its covariance (21,155.55) is far smaller than the housing covariance (33,491,250). This is the comparison covariance could not make. Squaring: SAT $r^2 \approx 0.88$, housing $r^2 \approx 0.75$.

7. Excel and Python

Goal	Excel	Result (housing)
Correlation coefficient (built in)	=CORREL(C6:C10, D6:D10)	0.8671
Same, alternative function	=PEARSON(C6:C10, D6:D10)	0.8671
Built by hand, as in the workbook	=G13/C12/D12	0.8671
Standard deviation (sample)	=STDEV.S(C6:C10)	222.47

```
import statistics as st
import numpy as np
import pandas as pd

size  = [650, 785, 1200, 720, 975]
price = [772000, 998000, 1200000, 800000, 895000]

st.correlation(size, price)          # 0.8671128624942824  (Python 3.10+)
np.corrcoef(size, price)[0, 1]       # 0.8671128624942823  (2 x 2 matrix: take [0, 1])
pd.Series(size).corr(pd.Series(price))  # 0.8671128624942824
```

Good to know
Because the $n - 1$ (or N) cancels out, the correlation is the same whether you use sample or population covariance and standard deviations, as long as you use them consistently. `np.corrcoef` returns a 2×2 matrix, so pick the off-diagonal entry `[0, 1]`.

8. Revision kit

Formula sheet

Quantity	Formula
Correlation coefficient	$r = \text{cov}(x, y) \div (s_x \cdot s_y)$
Sample covariance	$\Sigma(x - \bar{x})(y - \bar{y}) \div (n - 1)$
Sample standard deviation	$s = \sqrt{[\ \Sigma(x - \bar{x})^2 \div (n - 1)\ ]}$
Range	$-1 \le r \le 1$
Share explained (added)	r^2

Self-test (cover the answers and try first)

#	Question	Answer
1	Why do we use the correlation coefficient instead of the covariance?	Covariance's size is hard to interpret (units, scale). r has no units and lies between −1 and 1.
2	Between which values can r lie?	−1 to 1, including both ends. No value outside this range is possible.
3	What does r = 0.87 tell you about house size and price?	A strong positive linear relationship. It does **not** say prices rise by $870 per foot.
4	Is the correlation between x and y different from the correlation between y and x?	No, it is the same (the formula is symmetric).
5	Prices are converted from dollars to thousands of dollars. What happens to r?	Nothing. r is unchanged; only the covariance changes.
6	Two variables have r = 0. Are they necessarily independent?	No. r = 0 only rules out a *linear* relationship (see $y = x^2$).
7	Ice-cream sales and umbrella sales have r < 0. Does one cause the other?	No. Both follow the weather. Correlation does not imply causation.
8	Find r for x = 1, 2, 3 and y = 6, 4, 2.	The points lie on a falling straight line, so **r = −1**.

Common mistakes

- Reading r as a percentage or as a slope (“870 dollars per foot”).
- Concluding that correlation proves causation.
- Treating r = 0 as “completely unrelated” when the relationship may be curved.
- Mixing sample and population values of the covariance and the standard deviations in one formula.
- Judging strength from the covariance instead of from r.

9. Notes check: points to double-check

While preparing these notes I compared the class notes with the two workbooks. These points are worth correcting in your own copy; the numbers in this document follow the workbooks.

Where	What I noticed	Suggested fix
Class notes, range of r	“no way to find the correlation coefficient is ≥ 1 or ≤ 1” is the wrong way round and would exclude 1 and −1 as well.	Write: r is always between −1 and 1, inclusive (−1 ≤ r ≤ 1).
Class notes, r = 0.87	“for each foot increase in size its price rises to 870 dollars”	r is unit-free and cannot be read as dollars per foot. The slope is about $677 per ft. (cov ÷ variance of size). Say instead: strong positive correlation.
Class notes, 1 sq. ft. = 1000 Rs	Mixes rupees (Rs) and dollars ($) in the same example.	Use one currency, for example 1 sq. ft. = $1,000 and 5 sq. ft. = $5,000.
Class notes, “size only decides the price”	r = 0.87 is not 1, and $r^2 \approx 0.75$, so other factors also matter.	Say: size is the main factor, but not the only one.
Class notes, “r = 0 absolutely independent”	Independent variables have r = 0, but r = 0 does not by itself prove independence.	Say: r = 0 means no linear relationship.
Workbook labels	Size column is labelled “Size (ft.)”; the sample size (5) and the divisor (÷ 4) are typed into the cells.	Relabel as square feet (ft^2) if that is the meaning. Use =COUNT(...) and =G11/(G12-1) so they update.
03_Correlation_Example.xlsx	The “cov” and “Covariance2” sheets are the unchanged housing sheets from the previous lesson, headed “Housing data”.	Harmless; they are not used by the SAT solution. Delete them if you want a cleaner file.

In one line each

- **$r = \text{cov}(x, y) \div (s_x\, s_y)$**, a unit-free number between −1 and 1.
- Sign gives the direction; the distance from 0 gives the strength. **r = 0**: no linear link.
- Housing: r = **0.87**. SAT: r = **0.94**. Both strong and positive.
- **Correlation does not imply causation.**

Source: class notes (Part 6), *02_Correlation.xlsx* (attached on page 2) and *03_Correlation_Example.xlsx* (attached on page 6). The scatter gallery, the r^2 explanation, the scale and U-shape checks, the self-test and the notes check were added for clarity and are marked as additions. All numbers were checked by computation against the workbooks.